

GARFIELD classifies disease-relevant genomic features through integration of functional annotations with association signals

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Loci discovered by genome-wide association studies predominantly map outside protein-coding genes. The interpretation of the functional consequences of non-coding variants can be greatly enhanced by catalogs of regulatory genomic regions in cell lines and primary tissues. However, robust and readily applicable methods are still lacking by which to systematically evaluate the contribution of these regions to genetic variation implicated in diseases or quantitative traits. Here we propose a novel approach that leverages genome-wide association studies' findings with regulatory or functional annotations to classify features relevant to a phenotype of interest. Within our framework, we account for major sources of confounding not offered by current methods. We further assess enrichment of genome-wide association studies for 19 traits within Encyclopedia of DNA Elements- and Roadmap-derived regulatory regions. We characterize unique enrichment patterns for traits and annotations driving novel biological insights. The method is implemented in standalone software and an R package, to facilitate its application by the research community.

Genome-wide association studies (GWAS) in humans have uncovered susceptibility variants for complex diseases and biomedical quantitative traits, and more than 75,000 associations have been found to date^{1,2}, representing a large investment in resources, time and organization by the worldwide research community. The majority (~90%) of variants implicated are classified as either intronic or intergenic³ and thus cannot be readily assigned to an underlying cellular or molecular mechanism. This has prompted a number of efforts to annotate the putative functional consequences of variants in cell-specific contexts from experimentally derived regulatory regions (for example, regions marked by histone modifications or transcription-factor binding^{3–6}), principally as a means to inform and accelerate functional validation.

The robust identification of the combinations of annotations for these regulatory regions (henceforth referred to generically as 'regulatory annotations'), and cell types that are biologically most informative for a given disease or quantitative trait of interest (henceforth referred to generically as 'phenotype'), requires the ability to confidently distinguish correlations driven by biology from those arising by chance. Regulatory annotations may cover a large proportion of the genome, and thus many disease-associated variants will map within these by chance. In addition, the heterogeneous distribution of genetic variants and functional regions in the genome may result in their nonrandom association with genomic features such as genes^{7,8}, which in turn may drive spurious correlations that confound correct interpretation of these correlation patterns.

Functional enrichment methods assess the relative contribution of regulatory annotations to a phenotype of interest. In their simplest

implementation, these estimate enrichment of association *P* values (or *z* scores) on the basis of comparisons of the full set of genome-wide association (GWA) variants^{9–13} or on subsets of highly associated variants (for example, genome-wide-significant variants)^{14–16}. These approaches have identified many biologically plausible patterns of enrichment and can be used broadly for ranking the relative contribution of features. For instance, variants associated with lipid traits and Crohn's disease are enriched in open chromatin derived from liver and immune cells, respectively¹³, thereby reflecting biological functions. However, there is currently little confidence in interpreting unexpected enrichments, because of various statistical concerns. First, overly simplistic models that do not account for known confounders, such as local linkage disequilibrium (LD) and local gene density, can lead to spurious enrichment patterns¹⁴. Second, tests based on subsets of variants typically probe a limited number of genomic features, whereas evidence of enrichment occurs well below genome-wide significance^{11,12}. Owing to the large number of annotations now available, a third problem has emerged—that of prioritizing the most informative set from a large number of often-correlated functional annotations. Methodological improvements are thus needed to increase the accuracy of inference and to realize the full potential of those costly experiments in focused analysis.

Here we present a novel statistical approach that leverages GWAS findings with functional (that is, regulatory or protein-coding) annotations to find features relevant to a phenotype of interest. This method accounts for LD, matched genotyping variants and local gene density with the application of logistic regression to derive

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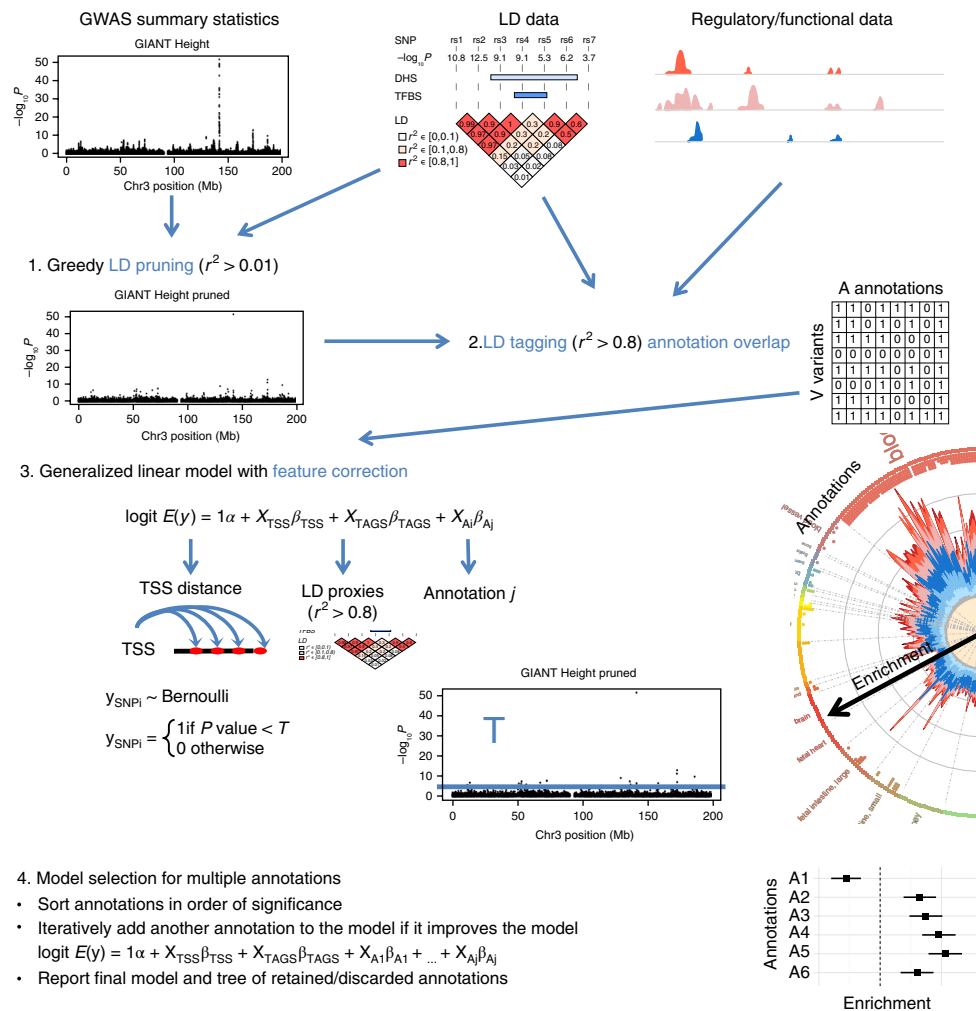


Fig. 1 | Outline of the GARFIELD method. Top, three inputs (annotations, P values and LD data) are used for the first two analytical steps (LD pruning (1) and variant functional annotation (2)), which result in a binary annotation overlap matrix of V pruned variants and A annotations. Middle, a logistic regression approach is used to test each annotation for enrichment at a GWAS significance P -value threshold T while controlling for confounding features such as TSS distance and number of LD proxies. Bottom, model-selection procedure for multiple annotations.

statistical significance. We named our method GARFIELD, which stands for GWAS analysis of regulatory or functional information enrichment with LD correction. We applied GARFIELD to analyze the enrichment patterns of publicly available GWAS summary statistics, using regulatory maps from the Encyclopedia of DNA Elements (ENCODE)³ and Roadmap Epigenomics⁵ projects. We describe expected and novel enrichments that illustrate the molecular and cellular basis of well-studied traits, which we expect will help drive novel biological insights and enhance efforts to prioritize variants for focused functional exploration. Finally, we developed new software to facilitate the application of our approach by the research community, as well as tools for effective visualization of enrichment results that scale to thousands of potential functional elements.

Results

Method overview. The analysis workflow implemented in GARFIELD is summarized in Fig. 1 and Methods. The method requires four inputs: (i) a set of genome-wide genetic variant association P values with a phenotype of interest; (ii) genome-wide genomic coordinates for regulatory annotations of interest; (iii) lists of LD tags for each variant ($r^2 \geq 0.01$ and $r^2 \geq 0.8$ within 1-Mb windows) from a reference population of interest (for example, Caucasian); and (iv) the distance of each variant to the nearest

transcription start site (TSS). Given these inputs, the method first uses a greedy procedure to extract a set of independent variants from the genome-wide genetic variants, using LD ($r^2 \geq 0.01$) and distance information ('LD pruning step'). Second, it annotates each variant with a regulatory annotation if either the variant or a correlated variant ($r^2 \geq 0.8$) overlaps the feature ('LD tagging annotation step'). Third, it calculates odds ratios and enrichment P values at different GWAS P -value thresholds (denoted T) for each annotation by using a logistic regression model with 'feature matching' (Methods) on variants by distance to the nearest TSS and number of LD proxies ($r^2 \geq 0.8$) (used as categorical covariates). This pruning strategy is conservative, because a potential loss of the true causal variant at a small fraction of the loci due to pruning will be offset by the analysis of genome-wide-enrichment patterns. We thus believe that this is a conservative but sound approach for identifying annotations that harbor more GWAS variants (at a given threshold, T) than would be expected by chance. To correct for multiple testing on the number of different annotations, we further estimate the effective number of independent annotations by using the eigenvalues of the correlation matrix of the binary annotation overlap matrix from Fig. 1 (adapted from Galwey¹⁷) (Methods) and then apply a Bonferroni correction at the 95% significance level. This procedure takes into account the tissue-selective components of regulatory data, namely that closely

related cell types and tissues are more similar to each other than to those not closely related. Our single annotation approach can be viewed as an extension of Maurano et al.¹¹ (Supplementary Table 1), with two critical improvements. First, we account for the effect of local variant correlations by restricting enrichment calculations to sets of independent variants (LD pruning step). Second, we use a testing procedure that accounts for systematic differences in gene distance and number of proxies in the variant set.

Additionally, we implement a heuristic procedure to combine the biological signal contained in correlated annotations, which allows us to identify conditionally independent sets of regulatory annotations underpinning the enrichment signals. To reduce the computational burden of searching through all possible combinations of available annotations, we first obtain enrichment *P* values for all annotations separately, by using the default single-annotation GARFIELD model. We then rank all statistically significant annotations by their enrichment *P* values and iteratively add each such annotation to the model if doing so significantly improves the model fit relative to the model lacking the annotation (analysis of deviance using a chi-square test).

We compared GARFIELD with five widely used alternative methods (LDSC¹⁰, fgwas¹³, GoShifter¹⁴, GREGOR¹⁶ and GPA⁹), while noting that benchmarking of methods is typically best done by investigators independent from the method developers. To estimate the false-positive rate (FPR), we used 21 real-disease or quantitative-trait GWASs with the required summary statistics for all methods and greater than five independent genetic variants at the $T < 10^{-8}$ threshold (Methods). We assessed the enrichment of each trait against 1,000 peak-region annotations, simulated to match observed peak lengths and between-peak distances for deoxyribonuclease 1 (DNase I)-hypersensitive sites (DHS) in HepG2 cells (ENCODE). We compared GARFIELD with the five alternative methods and to a naive model, in which enrichment odds ratios are calculated without accounting for LD or other features. FPRs were estimated by the observed proportion of significantly enriched annotations per phenotype (Methods). At the 5% significance level, methods not modeling LD produced significantly inflated FPRs (0.15 and 0.33 on average for naive and GPA, respectively) (Fig. 2a). GARFIELD, fgwas, LDSC and GoShifter preserved the FPR for all traits, whereas GREGOR yielded more false-positive results than expected (average FPR=0.09). Further assessment of GARFIELD for a set of 29 traits showed that FPRs were also preserved when lowering the threshold from $T < 10^{-8}$ to $T < 10^{-5}$ (Supplementary Fig. 1a).

To assess the value of feature matching in significance testing, we applied GARFIELD, with and without feature correction, to 424 open chromatin annotations in 29 phenotypes at the $T < 10^{-8}$ threshold. As expected, we found that feature matching controls for biases in enrichment analysis by significantly decreasing the number of observed enrichments (Wilcoxon signed-rank-test proportion median=0.46, $P=1.4 \times 10^{-4}$) (Fig. 2b). We further explored the relative contribution of each feature by comparing the number of significant enrichments detected in a feature-corrected model compared with the uncorrected model. We found median proportion reduction estimates of enrichment of 0.34 ($P=1.4 \times 10^{-4}$) and 0.10 ($P=1.1 \times 10^{-3}$) for the number of LD proxies and TSS distance, respectively (Supplementary Fig. 1b–c). Estimates were concordant between GWAS *P*-value thresholds (Supplementary Fig. 1d). These tests suggest that LD proxy number is the single most important confounder, although it is not sufficient to correct for individually when compared to the model correcting for both features together.

Enrichment in open chromatin regions. To assess the relative enrichment of phenotype-genotype associations in different cell types, we first applied GARFIELD to a generic regulatory annotation denoting open chromatin (DHS) in 424 cell lines and primary cell

types from ENCODE³ and Roadmap Epigenomics⁵ (Supplementary Table 2). We considered five diseases and 24 quantitative traits with publicly available GWAS summary statistics. For each trait-annotation pair, we derived enrichment estimates at eight GWAS *P*-value thresholds ($T < 10^{-1}$ to $T < 10^{-8}$). At the most stringent cutoff ($T < 10^{-8}$), there were a median of 21 independent variants per trait after LD pruning (range 0–117, Table 1 and Supplementary Table 3), whereas at a more permissive threshold ($T < 10^{-5}$), there were a median of 76 variants per trait (range 11–619).

We found statistically significant enrichments ($P < 2.6 \times 10^{-4}$; Methods, Supplementary Note) for most traits considered, thus highlighting clear differences in enrichment patterns among traits (Supplementary Table 4). As clearly visible from enrichment wheel plots, some traits displayed relatively ubiquitous enrichment (for example height, Fig. 3a), whereas others showed relatively narrow enrichment (for example ulcerative colitis, Fig. 3b; additional data in Supplementary Fig. 2). Blood cells were overall the most enriched tissue type in hematological traits and autoimmune diseases but provided little or no enrichment for glycemic, blood pressure and anthropometric traits (except height, which was enriched in nearly all tissues). As predicted, incorporating subthreshold associations ($T < 10^{-5}$) increased the resolution of enrichment patterns across traits (Table 1). For instance, at $T < 10^{-8}$, there were no annotations enriched for waist/hip ratio, whereas at $T < 10^{-5}$, there were 19 significant enrichments, 18 of which arose from muscle or fetal muscle tissue. For HbA1C and fasting glucose, again there were no enrichments at $T < 10^{-8}$, whereas at $T < 10^{-5}$, we uncovered links to blood, fetal stomach and fetal intestine tissues. Additionally, for low-density-lipoprotein cholesterol we found a single enrichment in colon at $T < 10^{-8}$, whereas the permissive threshold allowed us to detect a much higher number of relevant annotations (75), including liver, blood and fetal intestine cell types. Overall, 89% of the enrichments at the $T < 10^{-8}$ threshold were also identified at $T < 10^{-5}$ (between-threshold $\log_{10}$ enrichment *P*-value correlation=0.85) (Supplementary Fig. 3), showing a high degree of agreement between thresholds.

The observed enrichments reflect the current understanding of key cellular types for disease, augmented with novel observations. In the former category were enrichments of lipid traits in blood, liver, fetal intestine and fetal thymus cell types, of hematological traits in blood and of autoimmune diseases (ulcerative colitis, Crohn's disease and inflammatory bowel disease) in the blood and fetal intestine^{11,13,18}. Potentially interesting examples of the latter category include the enrichment of Caco-2 (a well-established gut epithelium cellular model) elements for low-density-lipoprotein cholesterol, the enrichment of (fetal) muscle and placenta cell-type elements in high-density-lipoprotein cholesterol and fetal intestine in hemoglobin. For each trait, we also used GARFIELD's heuristic multiple annotation approach illustrated earlier to further prioritize a parsimonious set of noncorrelated cell types among those with significant enrichment. Only a small proportion of enriched annotations detected under univariate settings was retained in the multiple annotation model (proportion median, 17%, range, 2–100%; median number of annotations retained, 2, range, 1–8; Table 1 and Supplementary Fig. 4). For instance, in regard to height, we narrowed down the annotations from 364 to 7 (2%). These findings are suggestive of a high degree in redundancy between annotations, while also highlighting that in most cases, biological enrichments are driven by more than a single annotation. For instance, for high-density-lipoprotein cholesterol, we obtained conditionally independent signals derived from blood and liver cell types.

Next, we sought to evaluate GARFIELD against alternative enrichment methods when considering empirical phenotypes and DHS data. We performed enrichment analysis for each of the 21 traits from the simulation study in each of the 424 cell types, using

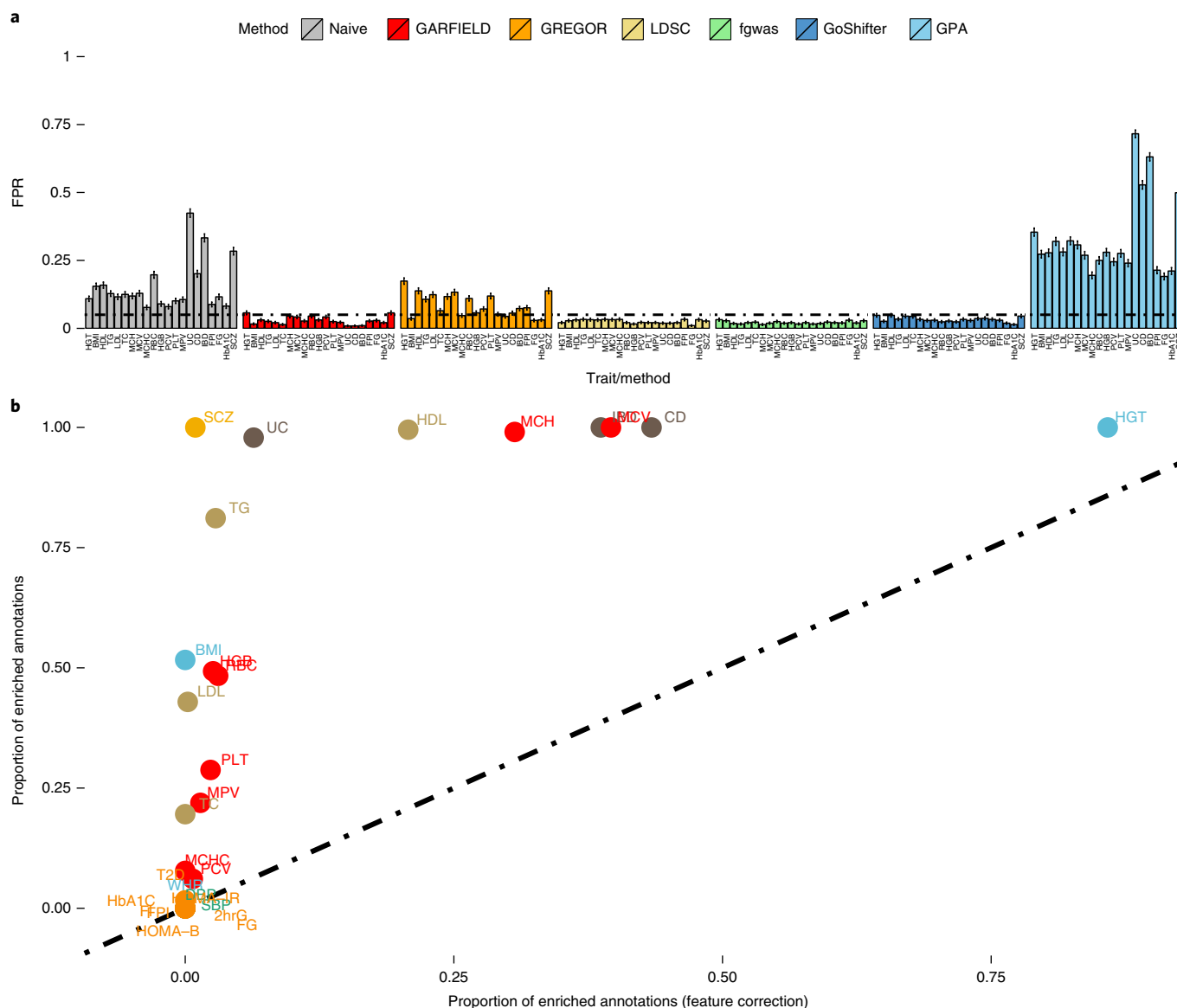


Fig. 2 | Method assessment. **a**, Estimated FPR from 21 publicly available disease or quantitative traits and $n=1,000$ simulated independent annotations. The broken black horizontal line denotes the 5% FPR threshold. Error bars denote standard errors. **b**, Comparison of the proportion of significant annotations (GARFIELD enrichment $P < 2.6 \times 10^{-4}$ for multiple-testing correction), found from models accounting for number of proxies and distance to nearest TSS, respectively (x axis), to a model not accounting for any feature (y axis), for each of 29 publicly available GWA studies and $n=424$ DHS annotations. The key for trait name labels is shown in Supplementary Table 3.

each of the five methods (GARFIELD, GoShifter, fgwas, GREGOR and LDSC) shown previously to preserve (or nearly preserve) FPR in simulations (Methods; Supplementary Table 5). GREGOR yielded the highest number of enrichments (median, 24, maximum, 398), followed by GARFIELD (median, 10, maximum, 364). fgwas and LDSC yielded intermediate levels of enrichment (median, 5, maximum, 327; median, 5, maximum, 144, respectively), whereas GoShifter was very conservative (median, 0, maximum, 5). Stratification of the enrichment to groups according to the number of methods supporting them further showed that GREGOR identified the largest number of enrichments found by at least one other method. GARFIELD closely followed GREGOR, whereas fgwas, LDSC and GoShifter showed much lower between-method concordance rates. Although GREGOR also identified the largest number of method-specific enrichments, the inflated FPR indicated that more enrichments were discovered at the cost of reporting more false positives, thus making the utility of GREGOR alone

less desirable in practice (Fig. 4a). In the absence of a truth set, the observation that GARFIELD captures a large proportion of enrichments consistent with other methods, while preserving the FPR, provides an indirect assessment of the robustness of our approach. Overall, enrichments of blood cell traits with blood cell regulatory annotations tended to be highly consistent among most methods (supported by GARFIELD for seven traits, GREGOR in eight, fgwas and LDSC in five and GoShifter in one; Fig. 4b), as was expected given their clear biological relevance. Likewise, we observed highly consistent results for height in most cell types: schizophrenia in blood and fetal brain; high-density-lipoprotein cholesterol in liver, blood and fetal placenta (supported by GARFIELD, GREGOR and fgwas); triglycerides in blood (GARFIELD, GREGOR, fgwas and LDSC); mean corpuscular volume in the fetal stomach, fetal spleen and fetal thymus; and mean corpuscular hemoglobin in fetal intestine, fetal stomach and fetal spleen, all of which were supported by at least three methods.

Table 1 | Summary of GARFIELD enrichment analyses in DHS, histone modifications and genomic segmentations per phenotype

Broad category	Similarity group	Trait	Abbreviation	Total independent SNPs genome wide	T < 10 ⁻⁸					T < 10 ⁻⁵					Top enriched categories						
					SNPs at threshold	Enriched DHS/cells	DHS/cells from multiple annotation approach	H3K27ac enrichment	H3K4me3 enrichment	Enriched seg. states/cells	Depleted seg. states/cells	SNPs at threshold	Enriched DHS/cells	DHS/cells from multiple annotation approach	H3K27ac enrichment	H3K4me3 enrichment	Enriched seg. states/cells	Depleted seg. states/cells	DHSs	Histone modifications	Segmentations
Anthropometric	1	Body mass index	BMI	50,166	16	0	-	0	1	14	0	62	0	-	0	0	14	0	Fetal muscle*	Connective/epithelial/bone	Active enhancers, tissues including adipose and skeletal muscle*
	1	Height	HGT	49,909	114	364	7	65	106	297	0	290	395	10	88	124	806	28			
	1	Waist/hip ratio adjusted for BMI	WHR	50,500	7	0	-	0	0	0	0	34	19	2	2	0	27	0			
Glycemic	2	2-h glucose adjusted for BMI	2HG	46,691	1	0	-	0	0	0	0	11	0	-	0	0	1	0	Blood and fetal intestine	Fetal intestine, spleen, adipose, and fetal pancreas	Poised promoters in pancreatic islets and stomach mucosa, active enhancers in pancreatic islets and transcription regulation in blood
	2	HbA1C	HbA1C	52,079	8	0	-	0	0	0	0	45	2	2	7	0	2	0			
	2	Fasting proinsulin	FPI	56,429	10	0	-	0	0	22	0	30	1	1	1	0	0	5			
	2	Fasting glucose	FG	51,578	13	0	-	0	0	0	0	43	11	3	1	2	7	0			
	2	Fasting insulin	FI	50,132	0	0	-	0	0	0	0	12	0	-	1	0	0	0			
	2	Type 2 diabetes	T2D	47,928	10	0	-	0	0	1	0	65	5	3	3	5	3	0			
	2	HOMA-IR	HOMA-IR	49,450	0	0	-	0	0	0	0	14	0	-	0	0	1	0			
	2	HOMA-B	HOMA-B	49,249	4	0	-	0	0	0	0	20	0	-	0	0	2	0			
	3	Diastolic blood pressure	DBP	52,446	7	0	-	0	0	0	0	43	0	-	1	0	2	0	Lung	Left ventricle	Transcription enhancers in blood and left ventricle
	3	Systolic blood pressure	SBP	52,592	7	0	-	0	0	4	0	48	1	1	3	0	9	0			
Lipids	4	High-density lipoprotein	HDL	61,981	66	88	3	22	6	135	26	150	154	3	65	35	483	119	Several (liver, blood, fetal intestine, colon, fetal thymus)	Liver, blood, gastrointestinal and adipose	Active enhancers in liver, transcription enhancers in fetal intestine and blood
	4	Total cholesterol	TC	62,029	64	0	-	2	1	8	6	131	15	3	20	16	122	14			
	4	Triglyceride	TG	61,999	36	12	2	6	1	2	0	95	43	3	29	32	41	0			
	4	Low-density lipoprotein	LDL	61,933	49	1	1	7	6	12	2	115	75	3	25	23	45	27			
Hematological	5	Hemoglobin count	HGB	49,658	21	11	3	10	2	141	0	84	20	3	15	4	97	0	Blood	HSC/blood/immune	Transcriptional regulation and active enhancers in blood
	5	Mean corpuscular volume	MCV	49,387	45	167	8	71	103	187	0	132	247	5	87	120	831	1			
Autoimmune disease	5	Red blood cell count	RBC	49,414	25	12	2	9	2	11	0	103	91	4	47	35	357	0			
	5	Mean corpuscular hemoglobin concentration	MCHC	49,466	13	0	-	2	0	3	0	45	6	2	3	0	10	0			
	5	Mean corpuscular hemoglobin	MCH	49,239	39	129	5	54	106	134	0	116	200	5	72	122	323	0			
	5	Packed cell volume	PCV	49,236	16	3	2	2	0	31	0	63	69	4	47	40	254	0			
	5	Mean platelet volume	MPV	56,683	27	6	2	3	1	4	0	76	18	3	17	3	37	1			
	5	Platelet count	PLT	58,181	35	10	2	15	9	19	0	111	66	4	74	95	352	2			
	6	Crohn's disease	CD	347,359	65	183	4	28	35	54	22	215	187	5	40	67	221	28	Blood, fetal intestine, fetal thymus	Blood immune and gastrointestinal cells	Active enhancers in blood immune cells and thymus
	6	Ulcerative colitis	UC	356,248	67	27	2	13	11	49	3	218	150	3	39	45	183	22			
	6	Inflammatory bowel disease	IBD	393,352	94	164	4	35	17	155	15	283	168	7	40	37	263	38			
	7	Schizophrenia	SCZ	170,825	117	4	1	17	15	98	0	619	2	2	8	1	29	6	Blood, fetal brain	CNS and HSC/blood/immune	Transcription/bivalent promoters in fetal brain

Total number of GWAS variants (n) per trait can be found in Supplementary Table 3.

*excluding height

Features, total independent SNPs significant in GWAS for a given T %excluding height

Total number of GWAS variants (n) per trait can be found in Supplementary Table 3

Columns denote (1) phenotype category and (2) its index; (3) trait full name and (4) abbreviation; (5) total number of variants after LD pruning; (6) number of independent SNPs at GWAS P-value threshold 10⁻⁸; (7) number of enriched cell types in open chromatin marks at GWAS P-value threshold 10⁻⁸; (8) number of conditionally independent cell types in open chromatin marks at GWAS P-value threshold 10⁻⁸; (9) number of enriched cell types in active enhancer H3K27ac at GWAS P-value threshold 10⁻⁸; (10) number of enriched cell types in active promoter H3K4me3 at GWAS P-value threshold 10⁻⁸; (11) number of enriched cell types/segmentation (seg.) states at GWAS P-value threshold 10⁻⁸; (12) number of depleted cell types/segmentation states at GWAS P-value threshold 10⁻⁸; (13) number of independent SNPs at GWAS P-value threshold 10⁻⁵; (14) number of enriched cell types in open chromatin marks at GWAS P-value threshold 10⁻⁵; (15) number of conditionally independent cell types in open chromatin marks at GWAS P-value threshold 10⁻⁵; (16) number of enriched cell types in H3K27ac at GWAS P-value threshold 10⁻⁵; (17) number of enriched cell types in H3K4me3 at GWAS P-value threshold 10⁻⁵; (18) number of enriched cell types/segmentation states at GWAS P-value threshold 10⁻⁵; (19) number of depleted cell types/segmentation states at GWAS P-value threshold 10⁻⁵; (20) tissues showing the largest enrichment in open chromatin states per trait category; and (22) tissues/segmentation states showing the largest enrichment per trait category, and (22) tissues/segmentation states showing the largest enrichment per trait category. Total number of GWAS variants (n) per trait can be found in Supplementary Table 3. - indicates no applicable data for testing.

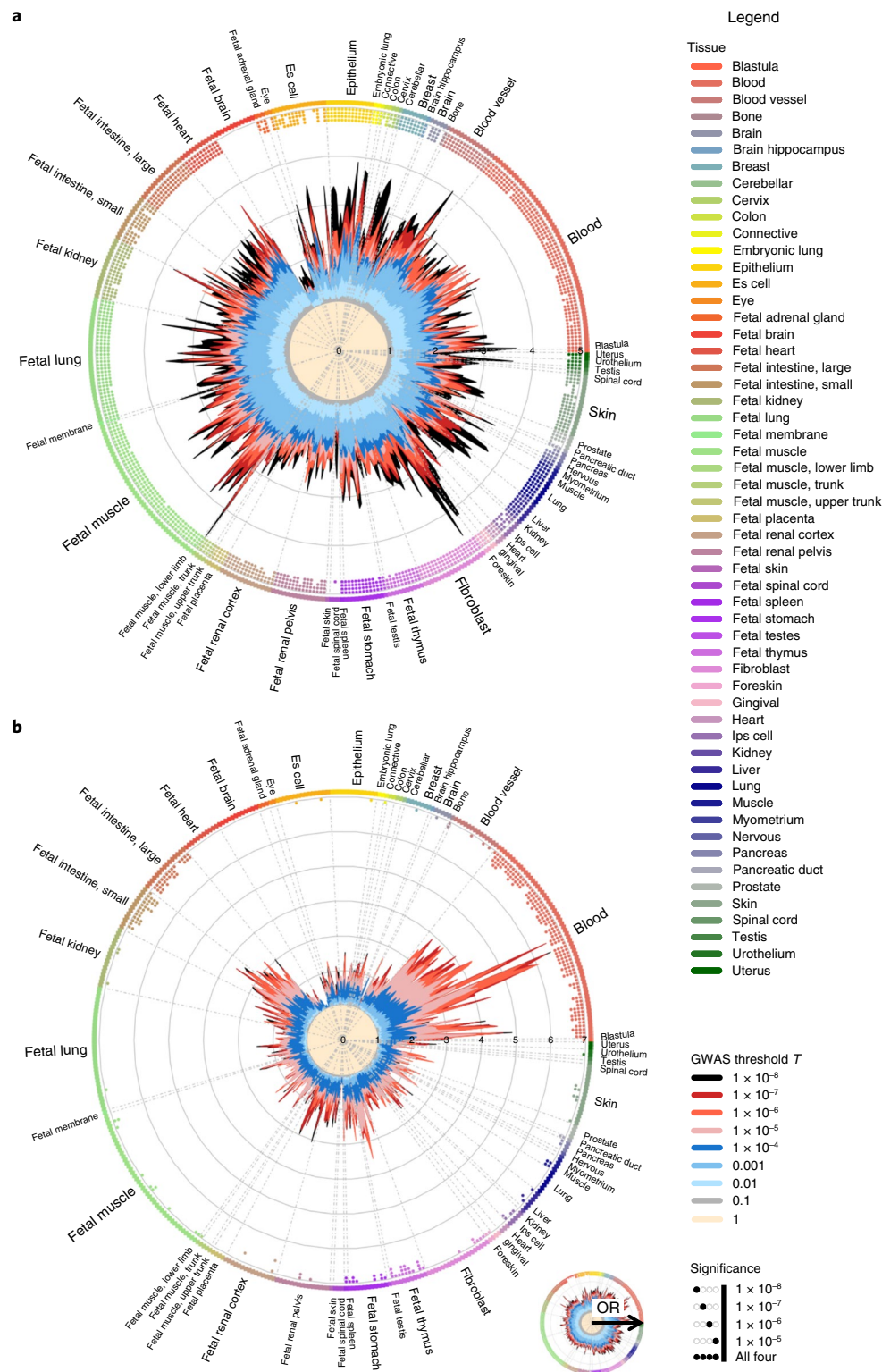


Fig. 3 | Enrichment of GWA analysis P values in DHS (hotspots). **a**, Height ($n=2,468,982$ GWAS variants). **b**, Ulcerative colitis ($n=11,113,952$ GWAS variants). Radial lines show odds ratio (OR) values at eight GWAS P -value thresholds (T) for all ENCODE and Roadmap Epigenomics DHS cell lines, sorted by tissue on the outer circle. Dots in the inner ring of the outer circle denote significant GARFIELD enrichment (if present) at $T < 10^{-5}$ (outermost) to $T < 10^{-8}$ (innermost) after multiple-testing correction for the number of effective annotations and are colored with respect to the tissue cell type tested (font size of tissue labels reflects the number of cell types from that tissue). Ulcerative colitis shows predominant enrichment in blood, fetal thymus and fetal intestine tissues, whereas height exhibits an overall enrichment. ES cell, embryonic stem cell; IPS cell, induced pluripotent stem cell.

Finally, we compared the average central processing unit (CPU) time used per method, trait and annotation, on the basis of the analysis of 21 traits and 424 annotations. GARFIELD was faster

than all other methods, with an average of 0 min 38 s required compared to 2 min 19 s for GoShifter, 6 min 42 s for LDSC, 16 min for fgwas and 0 min 58 s for GREGOR (Methods). It must be noted,

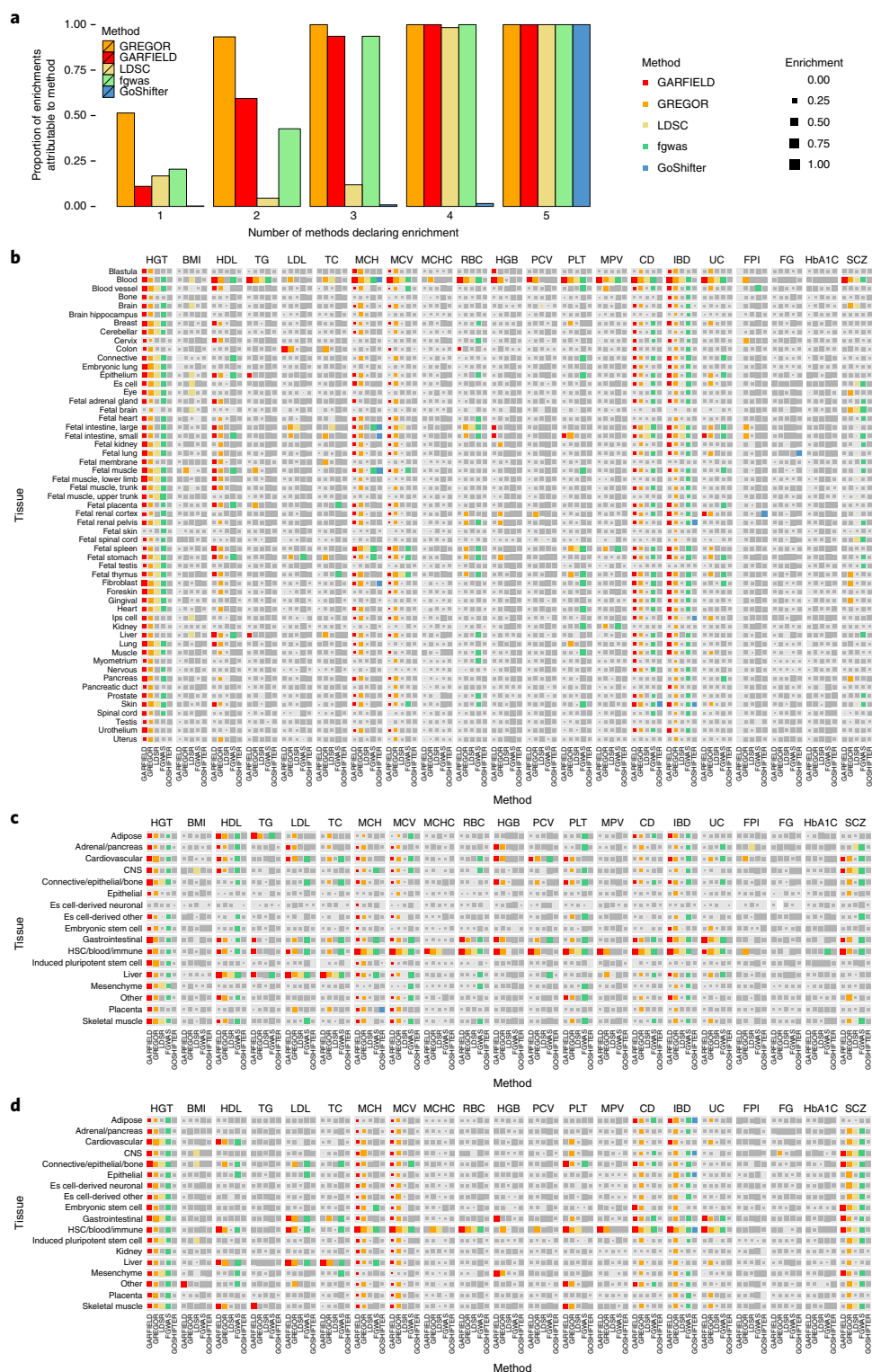


Fig. 4 | Method comparison for 21 GWAS datasets in DHS (hotspots) and histone modifications (H3K27ac and H3K4me3) at the $T < 10^{-8}$ GWAS significance threshold. **a, Proportion of enriched cell types in DHS identified by each method; enrichments are stratified by the number of methods supporting them. GARFIELD, fgwas and LDSC are restricted to positive enrichments only for those to be comparable to GREGOR and GoShifter. **b**, Summary of significant enrichments (per tissue and per method) for DHS data. A colored box denotes that the corresponding method found at least one significantly enriched cell type for that tissue after multiple-testing correction. Colors correspond to the different methods and are the same as in **a**. A gray box denotes that the enrichment did not reach significance. Additionally, the size of each box represents the relative magnitude of the enrichment. Because each method uses a different enrichment statistic, we scaled each of these separately per method and per trait (for example, for GARFIELD we scaled the odds ratios for all cell types for high-density lipoprotein so that 1 denotes the cell type with the highest enrichment found and 0 the lowest). **c**, Summary of significant enrichments (per tissue and per method) for H3K27ac data. **d**, Summary of significant enrichments (per tissue and per method) for H3K4me3 data. **b–d**, Sample sizes (n) per trait and trait name labels can be found in Supplementary Table 3, denoted by the number of variants in each GWAS study.**

however, that LDSC is fast to run but had the substantial computational burden of generating the necessary input files for our custom data (Supplementary Table 6).

Enrichment in promoter and enhancer marks. In light of the current knowledge of relevant links between cell types and complex traits, based on promoter and enhancer activity, we also sought to evaluate GARFIELD against alternative enrichment methods when considering empirical phenotypes and marks for the activity of H3K27ac and H3K4me3 in 127 cell types, similarly to DHS comparisons presented earlier.

We found statistically significant enrichments ($P < 5 \times 10^{-4}$; Methods) confirming known biology for both H3K27ac and H3K4me3 (Supplementary Table 5). In particular, height was enriched in most tissues for both regulatory marks; schizophrenia appeared predominantly enriched in central-nervous-system tissue, blood cell traits were enriched in HSC/blood/immune cell types, and lipid traits were enriched in liver tissues for both marks. The overall results also showed fewer and more specific enrichments in H3K27ac in comparison with H3K4me3 (mean 17, range 0–72, and mean 20, range 0–106 number of enrichments, respectively), in agreement with the higher cell-type specificity found in active enhancers versus active promoter regions.

Enrichment in genomic segmentations. We additionally sought to compare the relative enrichment of different types of functional genomic marks, by using ChromHMM¹⁵ data on genomic segmentations for 127 cell types (Supplementary Table 7). For each segmentation state and cell type, we analyzed our 29 phenotypes at two different GWAS P -value thresholds ($T < 10^{-5}$ and $T < 10^{-8}$). Overall, when considering only significantly enriched trait–annotation pairs ($P < 3.3 \times 10^{-5}$; Supplementary Table 8), we found higher levels of enrichment for promoters (median odds ratio, 3.4, range, 2.0–10.9 for $T < 10^{-5}$) and enhancers (median odds ratio, 3.8, range, 1.9–68.0) than for transcribed regions (median odds ratio, 2.6, range, 1.8–13.8) and depletion in quiescent regions (Fig. 5a) (similar patterns were obtained for $T < 10^{-8}$; Supplementary Fig. 5). Given that transcriptional states mainly mark active genes, the contrast of enrichment in transcriptional regions compared to the depletion in quiescent regions was unsurprising. Interestingly, the enhancer states consistently showed stronger enrichments than transcribed regions, an observation in agreement with enrichments of hematological traits in cell-matched regulatory states from the BLUEPRINT project¹⁹. To confirm these patterns, while controlling for the effect of annotation density on the number of enrichments found, we sought to compare only odds ratios for cell types enriched in both transcribed and enhancer pair states (and promoter and transcribed states). Similarly to our previous observations, the results showed on average higher odds ratios for enrichment for enhancers compared with transcribed regions (Fig. 5b) (with a similar but weaker effect for promoters), thus providing further evidence that our observation was not due to difference in power for enrichment detection between annotations of different density but was instead due to their biological relevance to the studied traits.

When considering cell-type specificity, again the trait height was the most ubiquitously enriched phenotype. In general, we found the highest odds ratios for anthropometric traits in active enhancers in adipose and skeletal muscle tissues; glycemic traits in active enhancers in pancreatic islets; poised promoters in pancreatic islets and stomach mucosa and transcription regulation in the blood; lipid traits in active enhancers in the liver, transcription enhancers in the blood and fetal intestine tissue; autoimmune diseases and blood traits in active enhancers in tissues including the blood and thymus; and psychiatric disorder in transcription and bivalent promoters in the fetal brain. As expected, incorporating subthreshold associations again greatly increased the resolution of enrichment patterns

across different traits (Table 1). For example, we found no significant enrichment at $T < 10^{-8}$ for the glycemic indices β -cell activity index (HOMA-B), glycosylated hemoglobin (HbA1C) and fasting glucose, whereas at $T < 10^{-5}$ HOMA-B was predominantly enriched in active enhancers in pancreatic islets and ES-13 cells, glycosylated hemoglobin in active enhancers in the psoas muscle and fasting glucose in poised promoters in pancreatic islets and stomach mucosa.

Finally, we assessed the extent to which traits shared significantly enriched annotations, by comparing the number of cell types per segmentation state that were found to be significantly enriched (or depleted) for a single trait compared with multiple traits (Fig. 5c and Supplementary Fig. 5). Our results confirmed patterns of higher cell-type specificity for enhancer states, with a median of 67% of cell types in enhancer states that were unique to a single trait compared with only 45% for promoter regions at $T < 10^{-5}$ (76 and 50%, respectively at $T < 10^{-8}$). These results confirm enhancer states as prime regions of interest¹⁹ when seeking to investigate gene function underlying complex trait and disease associations.

Software implementation. Many GWA studies seek to explore functional enrichment patterns but often rely on customized, in-house pipelines. We implemented GARFIELD as a standalone tool in C++ to facilitate its use by the research community (Methods). The software allows for enrichment analysis of any user-provided trait with variant GWAS P values and GRCh37 genomic coordinates. We provide more than 1,000 GENCODE²⁰, ENCODE³ and Roadmap Epigenomics⁵ precompiled annotations, UK10K sequence LD data and TSS distance information as a ready-to-use package. Furthermore, custom annotation data can easily be accommodated when provided in a simple bed format. In addition, we developed a Bioconductor package for the R statistical framework to further increase usability.

Discussion

Large-scale efforts^{3–6} have been devoted to systematically mapping molecular traits associated with genomic regulatory regions. These have greatly enhanced the annotation of putative functional consequences of noncoding variants in cell-specific contexts, and have further provided links to disease association. However, current methods that aim to evaluate the contribution of such regions to genetic variation in disease cannot always do so robustly or are not readily applicable for systematic analysis and comparison of broad sets of features. In particular, it has been shown that LD and gene density can confound enrichment analysis results¹⁴. Here we further estimated the relative effect of each of those features and identified LD as the largest confounder. Additionally, because of their design, different genotyping platforms (and imputation strategies) can create different biases (for example, number of variants and genomic location distribution). GARFIELD accounts for all those features, by obtaining independent signals and through expansion to relevant annotations by using a population-scale LD reference and feature matching. To the best of our knowledge, there is no other method that can do so without making extremely restrictive assumptions (for example, Pickrell¹³ assumes at most one causal variant at a given genomic region). Furthermore many available approaches use variants that reach genome-wide significance from association analysis ($T < 5 \times 10^{-8}$), although there is evidence of enrichment occurring well below that level^{11,12}. To capture these effects, GARFIELD allows for parallel enrichment analyses at multiple P -value subthresholds, thereby improving the power to detect statistically significant enrichment patterns by increasing the number of variants tested and enabling its application to traits with underpowered GWA studies. Finally, we provide a flexible software platform with effective visualization to enable researchers to carry out simultaneous enrichment analysis for thousands of annotations at multiple association thresholds.

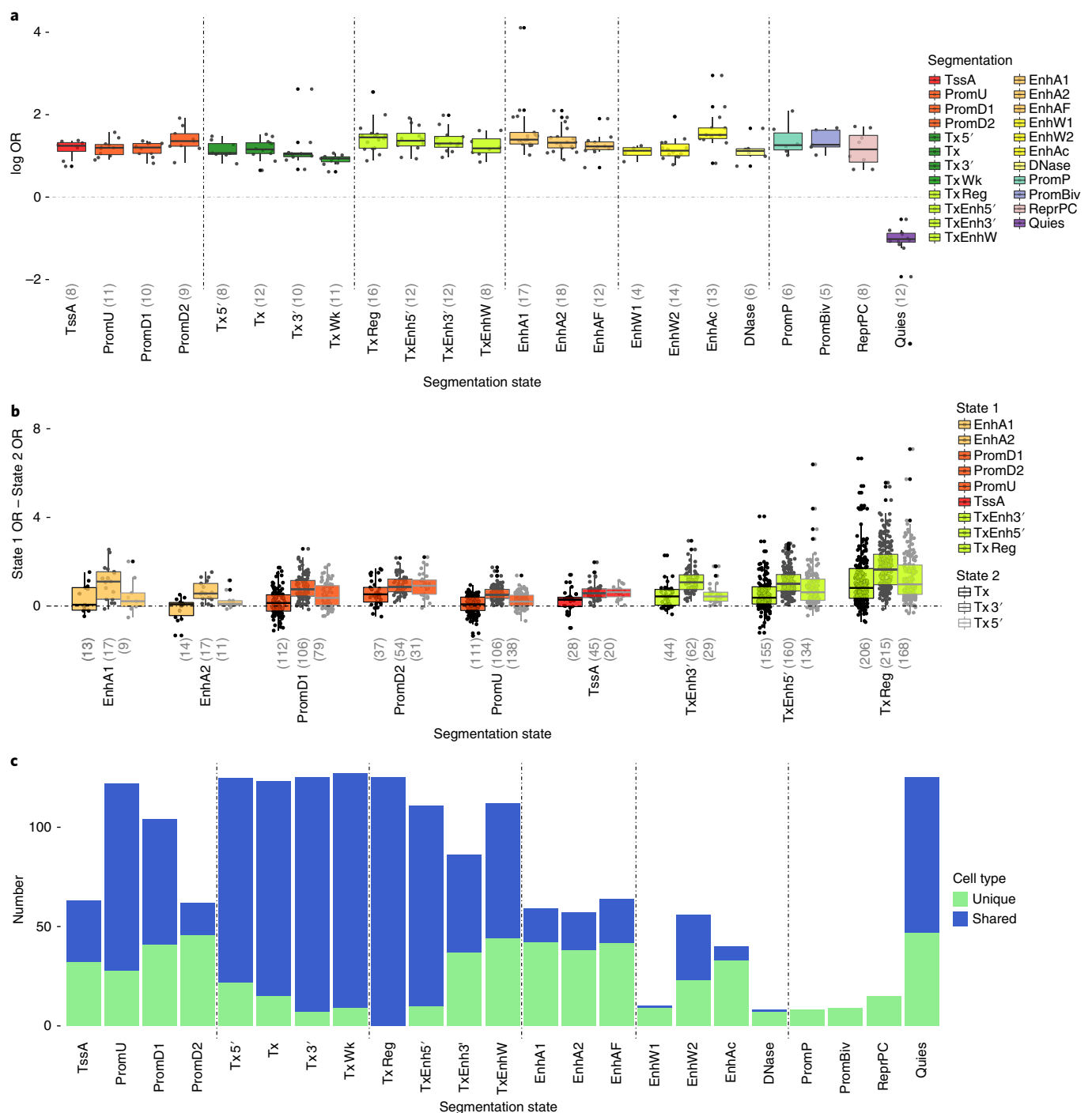


Fig. 5 | Enrichment levels (log OR) and extent of sharing between traits for 25-state chromatin segmentations of the National Institutes of Health Roadmap and ENCODE projects at the $T < 10^{-5}$ GWAS significance threshold. **a, Distribution of significant log OR values across the 29 traits considered, split by segmentation state and colored to highlight predicted functional elements (Supplementary Table 9). Number of points (n) is shown on the x axis below each category. **b**, Distribution of the pairwise difference between odds ratios from all enhancer, promoter and transcriptional enhancers and transcriptional regulatory states tested ('state 1') to odds ratios from transcription states for significant enrichments only ('state 2': for example, measuring odds ratios $OR_{EnhA1}^{ct} - OR_{Tx}^{ct}$ for all cell types (c) and traits (t) for which PP_{EnhA1}^{ct} and P_{Tx}^{ct} are both significant). Number of points (n) is shown on the x axis below each category. Box plots show the median (center line), upper and lower quartiles (box limits), whiskers, furthest point $< 1.5 \times$ interquartile range (whiskers), points in the distribution (gray points) and outliers (black points). **c**, Sharing of significantly enriched (or depleted) annotations ($n = 127$ cell types) across 27 phenotypes (excluding Crohn's disease and ulcerative colitis as categories of inflammatory bowel disease). The bar plot displays the number of cell types where an annotation is uniquely enriched/depleted in a trait or shared between traits.**

In our own application of GARFIELD on existing GWAS and functional datasets, we identified a broad set of largely expected or previously identified enrichments—for example, lipid traits in open

chromatin in liver, hematological traits in blood and anthropometric traits in active enhancers in adipose tissue. A number of GWAS hits did not show significant enrichments even with established cell

types when higher thresholds were used, but GARFIELD's stepwise, stratified approach uncovers these more nuanced enrichments, as shown in the case of pancreatic islets with fasting HOMA-B. By analyzing large-scale genome segmentation data, we assessed the relative contribution of each segmentation state to the phenotypic traits. We discovered a larger number of enrichments arising from transcription states as opposed to promoter and enhancer states, together with a larger number of shared cell types between traits. These findings may be biologically relevant or could equally be a result of statistically larger power for enrichment detection for broader region annotations. Here we show that study power differences are not responsible for higher odds-ratio values for significant enrichments in promoter and enhancer regions compared with transcribed regions, thus highlighting these as much more relevant for trait associated variants.

Robust, usable and modular methods are critical in the modern large-scale analysis arena, in which we expect many discoveries to come from principled combinations of heterogeneous datasets. In our hands, GARFIELD provided the greatest number of enrichments on real data among methods with full control of FPR in simulated data, and it was among the fastest methods. However, we acknowledge that, as authors of this method, we are not the group best positioned to provide unbiased benchmarking of these methods, and we look forward to independent analyses. We have already deployed GARFIELD in a number of association study settings, both in house and more broadly in the community. Our aim in developing it has been to provide the most robust statistical framework for analyzing functional enrichments coupled with practical ease of use and visualization, and we hope that the community will continue to exploit this tool to provide more insights into disease mechanisms.

URLs. Association summary statistics: Genetic Investigation of Anthropometric Traits consortium BMI²¹, height²² and waist/hip ratio adjusted for BMI²³, http://www.broadinstitute.org/collaboration/giant/index.php/GIANT_consortium; Meta-Analyses of Glucose and Insulin-related traits consortium BMI-adjusted 2-h glucose²⁴, HOMA-B, HOMA-IR, fasting glucose, fasting insulin²⁵, fasting proinsulin²⁶ and HbA1C²⁷, <http://www.magicinvestigators.org/downloads/>; Global lipid GWAS summary statistics for low-density lipoprotein, high-density lipoprotein, total cholesterol and triglyceride²⁸, <http://www.sph.umich.edu/csg/abecasis/public/lipids2010/>; International Inflammatory Bowel Disease Genetics Consortium data on Crohn's disease, ulcerative colitis and Inflammatory bowel disease²⁹, <http://www.ibdgenetics.org/downloads.html>; International Consortium for Blood Pressure consortium data on SBP and DBP³⁰, http://www.georgehretlab.org/icbp_088023401234-9812599.html; Diabetes Genetics Replication and Meta-Analyses type 2 diabetes³¹ GWAS summary statistics, <http://diagram-consortium.org/downloads.html>; Psychiatric Genomics Consortium data on schizophrenia, <http://www.med.unc.edu/pgc/>. DHS data: ENCODE DNase I hypersensitive sites, http://genome.ucsc.edu/cgi-bin/hgTrackUi?db=hg19&g=hub_4607_uniformDnase&hubUrl=http://ftp.ebi.ac.uk/pub/databases/ensembl/encode/integration_data_jan2011/hub.txt; National Institutes of Health Roadmap Epigenomics Mapping, http://www.genboree.org/EdaccData/Current-Release/experiment-sample/Chromatin_Accessibility/. H3K27ac, H3K4me3 and epigenome segmentation data: Processed NarrowPeak consolidated epigenome data and 25-state chromatin-state model on imputed data for 12 marks, http://egg2.wustl.edu/roadmap/web_portal/.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at <https://doi.org/10.1038/s41588-018-0322-6>.

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Author contributions

G.R.S.R., J.L.M., K.W., N.J.T., I.D. and N.S. contributed data or materials. E.B., G.R.S.R., I.D., J.L.M., N.S. and V.I. developed the method. V.I. analyzed the data. E.B., I.D., N.J.T., N.S. and V.I. provided critical interpretation of the results. M.G. and S.M. designed the tools. E.B., N.S. and V.I. wrote the manuscript. E.B., G.R.S.R., I.D., J.L.M., K.W., M.G., N.J.T., N.S., S.M. and V.I. evaluated the manuscript. E.B. and N.S. designed and managed the project.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Association summary statistics data. The GWAS summary statistics from the analysis of 29 disease and quantitative phenotypes were obtained from a number of sources (see URLs). From the Genetic Investigation of Anthropometric Traits consortium, we downloaded major studies on BMI²¹, height²² and waist/hip ratio adjusted for BMI²³. From the Meta-Analyses of Glucose and Insulin-related traits consortium, we downloaded data on BMI-adjusted 2-h glucose²⁴, HOMA-B, HOMA-IR, fasting glucose, fasting insulin²⁵, fasting proinsulin²⁶ and HbA1c²⁷. Global lipid GWAS summary statistics for low-density lipoprotein, high-density lipoprotein, total cholesterol and triglyceride were obtained from ref. ²⁸. Data on Crohn's disease, ulcerative colitis and inflammatory bowel disease²⁹ were obtained from the International Inflammatory Bowel Disease Genetics Consortium, and those for systolic blood pressure and diastolic blood pressure³⁰ were downloaded from the International Consortium for Blood Pressure consortium. Type 2 diabetes³¹ GWAS summary statistics were downloaded from Diabetes Genetics Replication and Meta-Analyses. Schizophrenia data from ref. ³² were obtained and analyzed. Blood trait data on hemoglobin count, mean corpuscular hemoglobin, mean corpuscular volume, mean corpuscular hemoglobin count, red blood cell count and packed cell volume were additionally obtained from van der Harst et al.³³; data on mean platelet volume and platelet count were obtained from Gieger et al.³⁴ (Supplementary Table 3).

DHS data. DNase I-hypersensitive sites (hotspots) were obtained from ENCODE and the NIH Roadmap Epigenomics Mapping (see URLs) for all available cell types. DHS data were processed according to the data-processing protocol described in an ENCODE study⁴. Further information on these data can be found in Supplementary Table 2.

H3K27ac and H3K4me3 data. Processed NarrowPeak consolidated epigenome data were downloaded from the Roadmap Epigenomics portal (see URLs) for all available cell types for H3K27ac and H3K4me3 marks (98 and 127 cell types, respectively). Cell-line information can be found in Supplementary Table 7.

Epigenome segmentation data. Data from a chromatin-state model with 25 states, based on imputed data for 12 marks (H3K4me1, H3K4me2, H3K4me3, H3K9ac, H3K27ac, H4K20me1, H3K79me2, H3K36me3, H3K9me3, H3K27me3, H2A.Z and DNase) across 111 Roadmap Epigenomics³⁵ and 16 ENCODE reference epigenomes, were downloaded from the Roadmap Epigenomics portal (see URLs). State and cell-line information can be found in Supplementary Tables 7 and 9.

LD data. LD information (proxies) was calculated by using PLINK³⁵ (v.1.7) and the --tag-r2 0.01 --tag-kb 500 (and --tag-r2 0.8 --tag-kb 500) flags to find all proxies within a 1-Mb window around each variant, at R^2 0.01 and 0.8. We computed these from the UK10K³⁶ sequence data on 3,621 samples from two population cohorts (TwinsUK and ALSPAC) (data described in ref. ³⁶). Variants not observed in the UK10K data were excluded from our analysis.

Data processing. Given a genome-wide distribution of P values for association with a given disease or quantitative trait, we performed the following preprocessing steps to calculate the level of enrichment and its significance for each annotation of interest. To remove possible biases due to either LD or dependence between variants, we computed the value of r^2 between all SNPs within 1-Mb windows, with values <0.01 between two variants being considered to denote (approximate) independence. Next, from the full set of genetic variants for each phenotype, we created an independent set of SNPs for which, to retain all possible GWAS signals, we sequentially found and retained the next-most-significant (lowest P value) variant independent of all other variants in our independence set. After LD pruning, an average of 2.2% (range 1.9–3.4%) of genome-wide variants remained in our independence set for enrichment analysis (Supplementary Table 3). Next, we annotated each independent SNP and considered it as overlapping a functional element if (i) the SNP itself resided in such a genomic region or (ii) at least one of its proxies in LD ($r^2 \geq 0.8$) and within 500 kb of it did. We included the latter because the association of a SNP in GWAS potentially tags the effect of other variants, which could underlie the observed association signal. The advantage of our greedy pruning over P -value-independent pruning is that we retained a larger proportion of potentially causal variants (or tags of such SNPs). This is particularly advantageous for GWA studies with low power and is more pronounced at more stringent pruning thresholds.

Quantifying enrichment and statistical significance. To determine the enrichment of GWAS signals within a given functional annotation at a genome-wide-significance P -value threshold T , we used the following logistic regression model:

$$\text{logit } E(y) = 1\alpha + X_{\text{TSS}}\beta_{\text{TSS}} + X_{\text{TAGS}}\beta_{\text{TAGS}} + X_{A_j}\beta_{A_j}$$

where $y_i = 1$ if SNP i has GWAS $P < T$, and $y_i = 0$ otherwise. The '1' denotes an intercept term (a vector of 1's), and X_{A_j} denotes a binary annotation covariate for annotation j . X_{TSS} and X_{TAGS} are categorical covariates denoting in which quantile bin of distance to nearest TSS and number of LD proxies ($r^2 \geq 0.8$) a variant falls (by default, we used five quantiles for TSS distance and 15 for number of LD proxies). These terms are added to account for possible biases in

the analysis due to correlation of the GWAS P -value distribution to them, which may also nonrandomly associate with functional data. Owing to the discreteness of the number of proxies and the skewness of their distribution in the pruned data, exact quantile binning is not always possible, in which case, we created a stepwise binning in which we iteratively found the first $(Q - q)$ th quantile from the remaining variants after having previously created q (out of Q) bins and removed those variants from consideration. We then calculated odds ratios and tested for their significance at $T = 10^{-1}$, 10^{-2} , ..., 10^{-8} for all traits at each given threshold.

Testing for significant association between an annotation and GWAS SNP status refers to testing for $\beta_{A_j} = 0$ versus $\beta_{A_j} \in \mathbb{R}$. If, additionally, $\beta_{A_j} > 0$, this denotes enrichment; otherwise, we consider it to be a depletion. The odds ratio statistic is then calculated via the following equation: $\beta_{A_j} = \log \text{OR}_{A_j}$.

Model selection for multiple annotations was done by (i) sorting annotations in order of significance from a single annotation model; (ii) iteratively trying to add an annotation to the model if this significantly improved the model fit ($P < 0.05$) given all other annotations in the model using the following model:

$$\text{logit } E(y) = 1\alpha + X_{\text{TSS}}\beta_{\text{TSS}} + X_{\text{TAGS}}\beta_{\text{TAGS}} + X_{A_1}\beta_{A_1} + \dots + X_{A_j}\beta_{A_j}$$

and (iii) reporting the final model and tree of retained/discarded annotations.

Multiple testing. To account for multiple testing in the number of annotations used, we applied a Bonferroni correction for the number of independent tests carried out. Owing to the nature of the data, annotations need not be (and are not in general) independent (for example, biological replicates of the same cell types). Thus, correcting for all annotations by assuming independence would be extremely stringent in practice. Instead, we estimated the effective number of independent tests performed (similarly to the method of Galwey¹⁷). More specifically, we took an independent subsample of SNPs and found the eigenvalues of the correlation matrix between all considered annotations, and then the effective number of independent tests from equation 16 in Galwey¹⁷. This procedure resulted in, at most, 194 independent annotations out of a total of 424 for the DHS data (for the 29 phenotypes considered), to which we applied Bonferroni correction ($P \approx 2.6 \times 10^{-4}$). Further details can be found in the Supplementary Note. Similarly, for the segmentation data a total of $25 \times 127 = 3,175$ annotations were used, which resulted in $P \approx 3.3 \times 10^{-5}$ after correcting for multiple testing on the effective number of independent annotations at the 5% significance level. Finally, for the histone modification data, we used a threshold of $P \approx 4.7 \times 10^{-4}$.

FPR. To obtain an estimate of the FPR for GARFIELD, we simulated 1,000 random annotations by mimicking the peak lengths and interpeak distances from the ENCODE HepG2 DHS cell line. We then performed enrichment analysis for each annotation–trait pair from the 1,000 simulated annotations and 29 publicly available disease or quantitative trait GWAS studies. We estimated FPR as the proportion of cell types showing significant enrichment for a given trait, and we further compared GARFIELD to each of six other tools for a subset of 21 traits with the necessary summary statistics for running all other approaches.

Analysis with other software. For method comparison analysis, we used a threshold of $T < 10^{-8}$ for GARFIELD, GREGOR and GoShifter and no threshold for fgwas and LDSC. Enrichment was defined as $P < 2.6 \times 10^{-4}$ and an effect with positive direction ($\text{OR}_{\text{GARFIELD}} > 1$, $\text{Enrichment}_{\text{LDSC}} > 1$, $\text{Estimate}_{\text{fgwas}} > 1$; GREGOR and GoShifter only test for enrichment and not for depletion, so these were used without this constraint).

fgwas. We used full GWAS summary statistics (no LD pruning or tagging) against each annotation at a time, as recommended by the fgwas user manual. Enrichment was defined by $P < 0.05$ for the FPR estimation and $P < 2.6 \times 10^{-4}$ for the real-data DHS analysis to correct for multiple testing.

LDSC. For each annotation, we prepared LD score (ldscore) files. For each annotation–trait pair, we then ran LD-score regression, accounting for the baseline model. We obtained enrichment P values based on the resulting regression coefficients, per the software documentation. Analysis was restricted to hapmap3 SNPs again, per the user-manual recommendation. Enrichment was defined by $P < 0.05$ for FPR estimation and $P < 2.6 \times 10^{-4}$ for the real-data DHS analysis to correct for multiple testing.

GoShifter. We restricted variants to those from the 1000 Genomes Project, owing to LD tagging in GoShifter when using the same panel. For each study, we selected variants with GWAS $P < 10^{-8}$ and pruned these similarly to GARFIELD according to LD $r^2 \geq 0.01$. Testing was done by using $r^2 \geq 0.8$ for LD tagging and 10,000 permutations. Enrichment was defined by $P < 0.05$ for FPR estimation and $P < 2.6 \times 10^{-4}$ for the real-data DHS analysis to correct for multiple testing. The P value of enrichment was calculated as the proportion of permutations producing overlap at least as extreme as the observed SNP annotation overlap.

GREGOR. We restricted the variants to those from the 1000 Genomes Project, owing to LD tagging in GREGOR when using the same panel. For each study, we selected variants with GWAS $P < 10^{-8}$ and pruned them similarly to GARFIELD according to LD $r^2 \geq 0.01$. Testing was done by using $r^2 \geq 0.8$ and 500 minimum neighboring SNPs for each tested variant. Enrichment was defined by $P < 0.05$ for

the FPR estimation and $P < 2.6 \times 10^{-4}$ for the real-data DHS analysis to correct for multiple testing.

GPA. We used full GWAS summary statistics, with no LD pruning or tagging, and used a maximum of 10,000 expectation-maximization (EM) iterations. Enrichment was defined by $P < 0.05$ and $q_1 > q_0$ for FPR estimation.

CPU time estimation. We compared total CPU usage times among different methods for the analyses of 21 traits and 424 annotations, and the respective average CPU times for a single trait–annotation pair. Analyses for each trait–annotation pair were run separately (whenever possible) on a compute cluster containing machines with the following architecture: Linux (x86-64) and 2×2.1 Ghz 16 core AMD 6378. We then recorded cumulative run time for all trait–annotation pairs and average run time for single trait–annotation pairs (Supplementary Table 6).

Segmentation odds-ratio distribution and between-trait sharing. From all significantly enriched cell types according to trait and segmentation state, we calculated the median odds ratio and then plotted its distribution (on a log scale) across traits to estimate the per-state odds ratio. Additionally, we took all significantly enriched cell types for pairs of annotations to remove the effect of power for annotation density and looked at the distribution of odds ratios for enhancer and promoter states versus those of transcription states. Finally, we counted the number of cell types per feature that were found to be significantly enriched (or depleted) in a single trait or shared between multiple traits.

Software. GARFIELD is available as both a standalone tool at <http://www.ebi.ac.uk/birney-srv/GARFIELD/> and an R package at <http://bioconductor.org/packages/release/bioc/html/garfield.html>. The tool consists of two main parts: (i) pruning and annotation of the GWA study of interest and (ii) calculating odds ratios and significance of the observed enrichment. Additionally, we provide scripts

for further prioritization of annotations by iteratively adding annotations in a joint model if these improve the model fit (chi-square test).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Code availability

Custom codes can be found at <http://www.ebi.ac.uk/birney-srv/GARFIELD/>.

Data availability

Web links for publicly available GWAS datasets and regulatory information databases are included in the URLs section. Restriction of availability applies to blood cell indices GWAS from van der Harst et al.³³ and Gieger et al.³⁴, which have been obtained through the manuscripts' authors. Any other data that support the findings of this study are available from the corresponding authors upon reasonable request.

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► Experimental design

1. Sample size

Describe how sample size was determined.

Not applicable. Our work is based on summary statistics, where we used the newest datasets available for a range of phenotypes.

2. Data exclusions

Describe any data exclusions.

Not applicable.
Done as part of individual published studies. For links to the data see Online Methods (page 10).

3. Replication

Describe whether the experimental findings were reliably reproduced.

Not applicable.
We introduce novel enrichment analysis method. Indirect replication of results is produced by the comparison to other methods.

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

Not applicable.
This was an observational study, thus no randomization was used for enrichment quantification.

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

Not applicable. We did not collect data for this study, but analyzed publicly available summary statistics.

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or the Methods section if additional space is needed).

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The <u>exact</u> sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.) |
| ☐ | ☒ | A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly. |
| ☐ | ☒ | A statement indicating how many times each experiment was replicated |
| ☐ | ☒ | The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section) |
| ☐ | ☒ | A description of any assumptions or corrections, such as an adjustment for multiple comparisons |
| ☐ | ☒ | The test results (e.g. p values) given as exact values whenever possible and with confidence intervals noted |
| ☐ | ☒ | A summary of the descriptive statistics, including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range) |
| ☒ | ☐ | Clearly defined error bars |

See the web collection on [statistics for biologists](#) for further resources and guidance.

► Software

Policy information about [availability of computer code](#)

7. Software

Describe the software used to analyze the data in this study.

We proposed novel enrichment analysis method GARFIELD which was mainly used for the analysis. An R-package for it was additionally developed for ease of use for the scientific community but was not used for this manuscript. A number of other tools have been used for comparison purposes (LDSC v1.0.0, GoShifter v0.2, GREGOR v1.2.1, fgwas v0.3.6, GPA v0.9.3) (see Online Methods).

For all studies, we encourage code deposition in a community repository (e.g. GitHub). Authors must make computer code available to editors and reviewers upon request. The *Nature Methods* [guidance for providing algorithms and software for publication](#) may be useful for any submission.

► Materials and reagents

Policy information about [availability of materials](#)

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.

Not applicable. No unique materials were used.

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).

Not applicable. No antibodies were used.

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.

Not applicable. No eukaryotic cell lines were used. All data comes from publicly available databases.

b. Describe the method of cell line authentication used.

Not applicable. No eukaryotic cell lines were used.

c. Report whether the cell lines were tested for mycoplasma contamination.

Not applicable. No eukaryotic cell lines were used.

d. If any of the cell lines used in the paper are listed in the database of commonly misidentified cell lines maintained by [ICLAC](#), provide a scientific rationale for their use.

Not applicable. No eukaryotic cell lines were used.

► Animals and human research participants

Policy information about [studies involving animals](#); when reporting animal research, follow the [ARRIVE guidelines](#)

11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.

Not applicable. No animals were used in this study.

Policy information about [studies involving human research participants](#)

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.

Not applicable. We are only using publicly available summary information from other studies. This study did not involve human research participants.